



**Rizal Technological University
COLLEGE OF EDUCATION
M. Eusebio Ave., Maybunga, Pasig City**

**Development of a Website System to Assess the Readiness of
Competencies of Pre-Service Teachers**

**Submitted to:
Instr. Alman P. Guaño**

**Submitted by:
Timbreza, Aljon M.
Navarro, Kirsten Arielle E.
Torio, Marvin B.
Zapeda, John Louie G.
Basijan, Angelo L
Legaspi, Jerick R.**

November 2025



Table of Contents

CHAPTER I	3
Introduction.....	3
Background of the Study	3
Statement of the Problem	4
Significance of the Study.....	4
Scope and Limitation of the Study	5
Theoretical Framework.....	6
Conceptual Framework	7
Definition of terms	8
Hypothesis.....	10
CHAPTER II	11
READINESS AND COMPETENCE OF PRE-SERVICE	11
PPST as the Foundation of Teacher Quality Benchmarking.....	11
Technology-Enabled Assessment Systems.....	12
Artificial Intelligence (AI) in Teacher Competency Evaluation.....	12
Synthesis	12
CHAPTER III	13
Research Design.....	13
Participants and Respondents	13
Materials and Instruments.....	14
Data Gathering Procedures	14
Statistical Treatment	15
REFERENCES:	17



CHAPTER I THE PROBLEM AND ITS BACKGROUND

Introduction

Teacher quality is a critical determinant of student learning outcomes, and ensuring that pre-service teachers develop the necessary competencies is essential for improving education systems. In the Philippines, the Education Commission Report (EDCOM Report) has highlighted challenges in teacher preparation, emphasizing the need to assess the effectiveness of teacher education programs. It highlights that many pre-service teachers enter the workforce with gaps in pedagogical skills, subject mastery, and classroom management, raising concerns about the alignment between teacher training and the actual demands of teaching.

Background of the Study

The preparation of pre-service teachers is a critical component of ensuring the quality of education in the Philippines. However, research suggests that many pre-service teachers feel unprepared for the demands of teaching, particularly in areas like classroom management, lesson planning, and assessment (De Las Penas et al., 2019). This study focuses on investigating the pre-service readiness and competency of student teachers in a Philippine university.

The school administration implemented various measures to enhance professional growth of pre service teacher including, facilitate access to instructional resources and foster a supportive work environment, thus teacher mentorship that imparted their experiences and knowledge, and provide practical ideas and guidance additionally the feedback and encouragement of the students can also play a big role in shaping pre service teacher. Through giving feedback particularly in instructional practices.

The study aims to bridge the needed quality of a pre-service teacher particularly in the aspect of readiness and competencies of individuals, thus, a significant disconnect between theoretical knowledge and hands-on practical application. The gaps can leave the pre-service teachers feeling unprepared and overextended, hindering their ability to effectively deliver the lessons and engage in a professional experience and meet the needs of future students.



The primary goal of this research is to assess the pre-service teachers, pertaining in the way of submitting their output, particularly the lesson plans. Mentor feedback, self-reflection is also included thus coming up with a website that can assess their artifacts of teaching like lesson plans, an AI is integrated in the website in order to examine the work if it's AI generated or self-hard work, the website also have test exams, pertaining about readiness of competency.

Statement of the Problem

The study, “Development of a Website System to Assess the Readiness and Competencies of Pre-Service Teachers,” aims to develop and evaluate a digital assessment platform designed to measure the readiness and competencies of BTVTED PST students. Grounded in the concerns raised by the EDCOM Report regarding gaps in pedagogical preparation, subject mastery, and classroom management among pre-service teachers in the Philippines, this study seeks to determine how a website-based assessment system can support teacher education institutions in monitoring and improving pre-service teacher readiness.

Specifically, the study aims to answer the following questions:

1. What are the key features of the developed website system designed to assess the readiness and competencies of pre-service teachers?
2. What is the level of acceptance of the developed website system among teachers, supervisors, and pre-service teachers based on the constructs of the Technology Acceptance Model (TAM)?
3. Is there a significant difference in the behavior or performance of pre-service teachers who have access to the developed website system compared to those who do not have access?
4. What challenges do teachers, supervisors, and pre-service teachers encounter before and during the use of the developed website system for assessing pre-service teacher readiness and competencies?

Significance of the Study

This study aims to introduce a website to assess the readiness of Bachelor of Technical-Vocational Teacher Education (BTVTED) Pre-service teachers on



their internship to their cooperating schools. The outcomes of this study will benefit multiple groups in the field of education:

- **Pre-Service Teacher.** This study helps pre-service teachers in identifying insights into their personal levels of readiness and teaching competencies based on the evaluation of data.
- **Cooperating Schools and Mentor Teachers.** The information gathered can help school partners and mentor teachers recognize the specific needs of Pre-Service Teachers. Teachers will be able to provide better direction and help during practice teaching as a result.
- **Educational Institutions.** This study can help DepEd and CHED improve the quality of technical-vocational education by making sure that new teachers have the abilities needed to fulfill 21st-century teaching requirements.
- **Future Research.** This study can be a reference for future researchers who want to conduct this kind of study such as a website about readiness of pre-service teachers.

Scope and Limitation of the Study

The study focuses on the development and implementation of a web based system intended to assess the readiness and competencies of pre-service teachers prior to their internship or teaching deployment. The system aims to evaluate various domains of teacher readiness such as content knowledge, pedagogical skills, classroom management, communication abilities, and professional values.

The student and faculty population from the College of Education at the partner institution will be recruited to participate in testing and system validation. The evaluation for this system will be relevant guidelines for evaluating software and measures for functionality, usability, and efficiency, and reliability.

This study will only measure pre-service teachers readiness using competencies established by the institution's established framework or standards and consider no other factors such as emotional readiness for the practicum or performance on an external practicum or future teaching impact.

Data collection and testing will limit participants to pre-service teacher and faculty positions at the selected institution and the generalizability of results to other colleges or universities may not be total. The accuracy of assessment



results will depend on the level of honesty in the responses and the reliability of the competency criteria provided by the institution.

Theoretical Framework

This study is anchored on several theories and models that support both the educational and technological dimensions of the study. It integrates educational standards, instructional design principles, and system development theories to meet technical functionality and align pre-service teachers with teacher education outcomes.

The Philippine Professional Standards for Teachers was established by the Department of Education, which serves as the primary foundation of this study. It thoroughly defines the competencies, domains, and indicators that pre-service teachers must possess to characterize effective teaching practices. In addition to that, it provides a clear continuum of teacher development from pre-service to in-service. The Philippine Professional Standards for Teachers serves as the basis for identifying and measuring the competencies and readiness indicators that will be assessed through the system that the researchers will produce.

The ADDIE Model, stands for Analysis, Design, Development, Implementation, and Evaluation is an instructional design framework that provides a systematic process to ensure that developed website possess instructional materials and digital learning tools that are consistent, effective, and efficient in identifying pre-service teachers needs and goals and evaluating instructional outcomes. It is widely used in developing educational tools, training programs, and digital learning systems. In the context of web development, the ADDIE model serves as the main theoretical foundation for developing a website system that assesses the readiness and competencies of pre-service teachers.

The Technology Acceptance Model or TAM is one of the most widely used theoretical models in studies related to adapting technology and information systems as it provides a clear definition of the factors that influence the users' attitudes toward using new technological tools or systems. It serves as the guiding theory for how pre-service teachers will accept and use the developed website system for assessing their competencies and readiness as part of their professional preparation based on the website's perceived usefulness and perceived ease of use. TAM helps ensure that the system



design prioritizes both technological efficiency and user-centered functionality, thereby enhancing the learning experience and readiness of future educators.

Conceptual Framework

This study integrates the Philippine Professional Standards for Teachers, ADDIE Model, and the Technology Acceptance Model to guide the development of a website system designed to assess the competency and readiness of pre-service teachers.

The Philippine Professional Standards for Teachers provides the basis for identifying and categorizing the domain indicators for competencies and readiness of pre-service teachers. These indicators define the expected skills, knowledge, and attitudes of the pre-service teachers that they must demonstrate to ensure the quality and effectiveness of teaching.

The ADDIE model supports the life cycle and process of website development. The Analysis phase ensures alignment with Philippine Professional Standards for Teachers. The Design and Development phases focus on creating the website's structure, interface, and assessment components to guarantee accurate and efficient evaluation, and functional and interactive platform for pre-service teachers. The Implementation phase involves the deployment of the system among pre-service teachers, while the Evaluation phase validates system effectiveness and identifies areas for improvement.

Lastly, the Technology Acceptance Model (TAM) determines the acceptance and usability of the developed system based- and guided by the constructs of Perceived Usefulness and Perceived Ease of Use by the pre-service teachers. These constructs influence the users' attitudes toward using the developed website and their behavioral intention to adopt it as part of their professional preparation. It ensures that the system not only functions effectively but is also perceived by pre-service teachers as a practical and accessible tool for self-assessment and skill development.

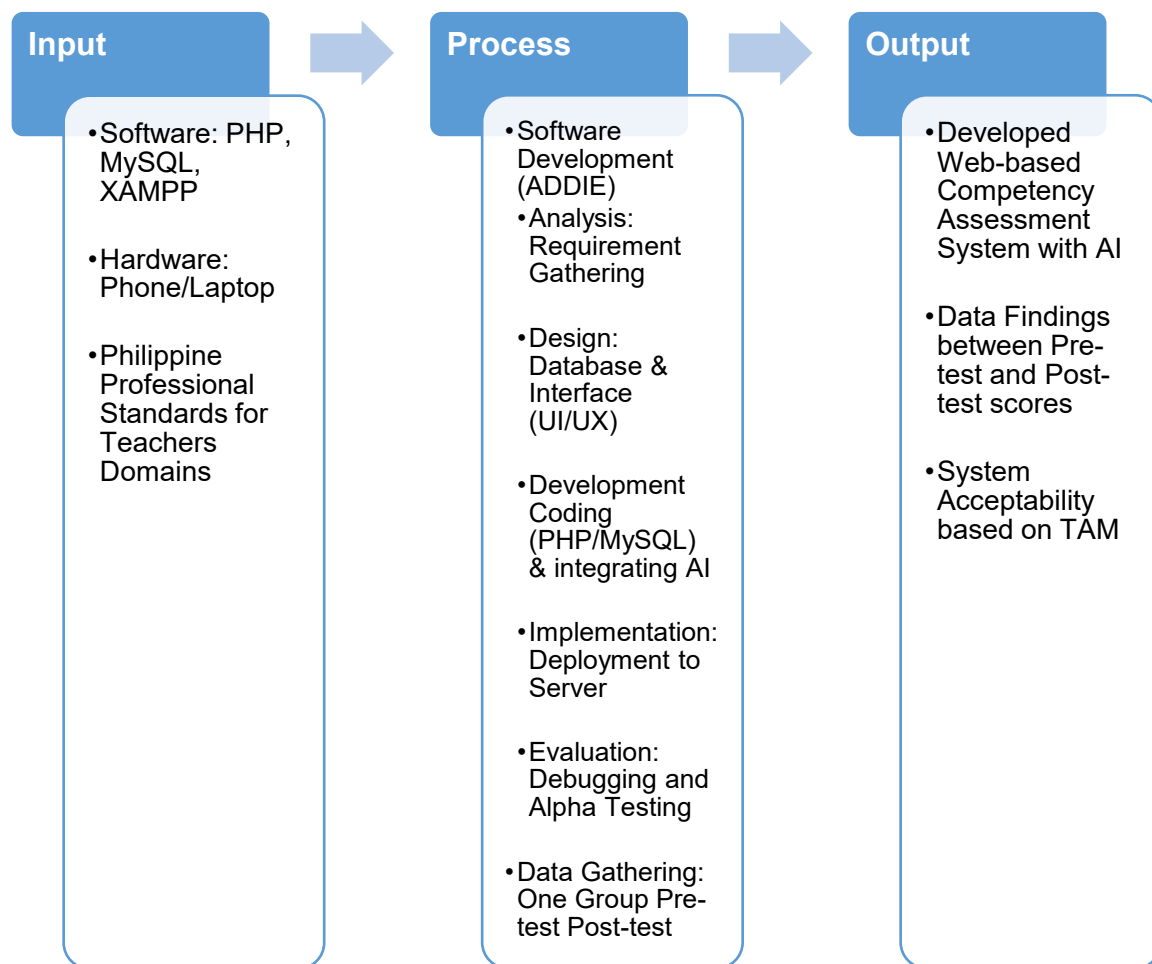


Figure 1: Research Paradigm of the Study

Definition of terms

Administrator - Authorized user who manages the website system and user accounts for system access, and maintains the data of the website system to ensure its integrity.

Assessment - The process of collecting and interpreting information regarding learners knowledge, skills, and attitudes.

BTVTED – Bachelor of Technical Vocation Teachers' Education



Competencies - This study refers to teaching skills and related attributes that indicate the readiness of the pre-service teacher for a real classroom teaching experience.

Database - The purposes of this study is used to user profiles, assessment results, and reports of pre-service teachers' competencies.

Evaluation - The evaluation of the developed website system in functionality, usability, efficiency, and reliability.

Faculty Evaluator - The teaching profession that evaluates and grades student performance and competencies.

Pre-Service Teachers - For the purposes of this study are the main users of the website system who will undergo an evaluation on their readiness prior to deployment for practice teaching.

Readiness - It refers to being fully available or prepared to carry out a defined task or responsibility. Readiness in this study is defined as the level of Preparedness of pre-service teachers with regard to their competencies and teaching performance prior to participating in an internship.

User Interface (UI) - The UI will be defined as the design and layout of the website that facilitates user engagement with the features of the system.

Website System - This study is a website system that was developed to provide assessments, record data, and generate reports about pre-service teachers competency readiness.



Hypothesis

Alternative Hypothesis (H1): Pre-service teachers working with the CORE developed website and utilizing its features will achieve an increment on their readiness and competency in teaching than those pre-service teachers who do not use the Core developed website.

Null Hypothesis (H0): There is no difference in readiness and competency of pre-service teachers that use CORE developed websites and those who do not use the CORE.



CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter constitutes the related literature and studies made regarding the Assessing readiness and competence of BTVTED PST to teach in their cooperating school which are imperative as to give a general overview of the relationship between these variables and what other research and study found about this. The first part of the chapter discussed the concepts of the variables while the second will focus on the importance and the relationship of the variables. To further explain our research topic, this part is composed of studies and works focused on:

READINESS AND COMPETENCE OF PRE-SERVICE

The study investigates the readiness and competence of new primary school teachers. It identifies their professional teaching readiness and analyzes their teaching competence using a mixed-method research design, incorporating qualitative interviews and observations alongside Qualitative surveys that the data were collected from various sources, including lecturers and graduate teachers. Findings reveal that most graduates feel prepared to enter the teaching profession, though they face challenges regarding welfare and professional legitimacy and must pursue further education. The foundation laid by their education contributes significantly to their competencies and overall readiness for professional teaching careers.

In addition the purpose is to provide the context of teachers readiness in many aspects including the physical preparedness, the body way of reacting when it comes to teaching, second is the psychological readiness that accommodates with the changing education system and lastly the cognitive readiness that encompasses the teachers way of thinking and how are they going to teach the subject effectively

PPST as the Foundation of Teacher Quality Benchmarking

The Philippine Professional Standards for Teachers (PPST) is the official national framework for defining teacher quality and progression (DepEd, 2017). It provides seven domains of competence—most critically in Domains 1 (Content Knowledge & Pedagogy), 2 (Learning Environments), and 7 (Professional Growth & Ethics). Research confirms that schools and SUCs still vary in actual alignment between teacher education outputs and PPST



indicators, showing a disconnect between policy and practice (Del Mundo, 2022).

Technology-Enabled Assessment Systems

The rapid shift to digital learning during the pandemic accelerated the adoption of web-based evaluation tools in teacher education (Al-Fraihat et al., 2020). Studies show that website-based digital portfolios and competency dashboards improve authenticity of performance assessment while increasing efficiency in documentation and feedback (Durak & Ataizi, 2021). Moreover, digital assessment systems based on the Technology Acceptance Model (TAM) are accepted by faculty and students as long as perceived usefulness and ease of use are high (Davis, 1989).

Artificial Intelligence (AI) in Teacher Competency Evaluation

AI-driven systems are now being integrated in teacher preparation—particularly in plagiarism detection, automated lesson plan review, and adaptive competency testing (Dwivedi et al., 2023). Pre-service teachers show high willingness to use AI tools, but moderate concern about fairness and authenticity (Cabanillas et al., 2024). AI's greatest strength lies not in replacing human evaluators—but in providing instant formative feedback, freeing mentors to focus on higher-order reflection rather than mechanical errors.

Synthesis

Existing studies collectively affirm the need for a centralized, PPST-aligned, AI-supported, web-based assessment system to track pre-service teacher competencies. Current evaluation systems are either manual, fragmented, or lacking AI-assisted analysis. Thus, the proposed website addresses three national gaps:

- 1) asynchronous digital assessment and feedback,
- 2) authentic competency evidence tracking, and
- 3) alignment with PPST and TAM for wide institutional acceptance.



CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the research method to be used, population and sampling scheme, description of the respondents, instruments used, data gathering procedures and the statistical treatment of the data.

Research Design

This study will utilize a mixed-method research design to assess the competencies and readiness of the BTVTEd pre-service teachers (PSTs) as evaluated by themselves, their cooperating teachers, and their supervisors. The survey questionnaire will analyze respondent's perception and level of acceptance of the developed website through product rating scales and scheduled interview. The achievement and attitude tests will analyze respondent's performance in the teaching environment as evaluated by themselves, their cooperating teachers and their supervisors to assess their performance through observation forms. Furthermore, this study will examine the level of agreement and discrepancies between the self-evaluations of pre-service teachers and the assessments provided by their cooperating teachers and supervisors.

Participants and Respondents

The participants of this study will consist of forty (40) Bachelor of Technical-Vocational Teacher Education (BTVTEd) pre-service teachers (PSTs) from Schools Division Offices of Mandaluyong and Pasig. The respondents will be divided into two main categories: those who will use the developed website and those who will not. This grouping is intended to compare the competencies and readiness of PSTs with and without the intervention of the developed system.

To ensure fair representation, the study will employ a Cluster Random Sampling technique. The total population of PSTs from each division will be divided into clusters according to their access to the developed website, which indicates whether they have laptops or not. In each division, there will be two clusters: (a) PSTs with developed websites, or those who have laptops and will use the website, and (b) PSTs without developed websites, or those who do not have laptops and will not use the website. From each of these clusters, ten (10)



respondents will be randomly selected to represent their group for both Mandaluyong and Pasig.

Materials and Instruments

This study will utilize a combination of quantitative and qualitative research instruments appropriate for a mixed-method research design. These instruments are designed to gather data on the competencies and readiness of BTVTEd pre-service teachers (PSTs), as evaluated by themselves, their cooperating teachers, and their supervisors. The instruments are survey questionnaires to analyze the level of acceptance, usability, and effectiveness of the developed website used to measure the competencies and readiness of PSTs. The achievement and attitude tests will measure the pre-service teachers' level of learning and teaching performance and their confidence in teaching and performing related tasks. The observation form will be used by cooperating teachers and supervisors to rate the actual classroom performance of pre-service teachers during their practicum. It will follow the domains set on the Philippine Professional Standards for Teachers. Lastly, the interview guide will be used to gather qualitative data on the perceptions, experiences, and suggestions of respondents regarding the use of the developed website and their evaluation process.

Data Gathering Procedures

The study will employ a systematic data gathering procedure divided into three distinct phases to ensure the collection of valid and reliable information relevant to the competencies and readiness of BTVTEd pre-service teachers and evaluation of developed website.

Phase 1: Pre-implementation and Pre-assessment

As this study utilizes a mixed-method research design, both quantitative and qualitative instruments will be used. Respondents will be oriented on the study's purpose, procedures, and use of the developed website. Quantitative data will be collected through survey questionnaires to measure website acceptance, usability, and effectiveness, as well as through achievement and attitude tests to evaluate PSTs' learning, teaching performance, and confidence in teaching.



Phase 2: System Intervention

The developed system, built using PHP and MySQL via XAMPP, will be deployed to the respondents. The respondents will use the website during their internship preparations. They will upload their lesson plans and take assessments within the system. The system's AI module will provide preliminary analysis on their inputs and checking for content.

Phase 3: Post-implementation and Evaluation

Cooperating teachers and supervisors will use observation forms to assess the pre-service teachers' classroom performance and professional behavior during their practicum. The respondents will answer the same set of competency assessments. The results will be compared against their pre-test scores to measure improvement. Respondents along with their cooperating teachers will rate the website using the Technology Acceptance Model survey to assess its Perceived Usefulness and Perceived Ease of Use.

All data will be systematically organized, coded, and analyzed using statistical methods for quantitative data and thematic analysis for qualitative data, providing a comprehensive understanding of pre-service teachers' competencies and readiness.

Statistical Treatment

To analyze the data effectively and test the hypothesis, the following statistical tools will be applied as follows:

Weighted Mean and Standard Deviation

These descriptive statistics will be used to analyze the survey questionnaire responses regarding the level of acceptance of the website based on the Technology Acceptance Model to describe the respondents' level of acceptance and overall perception of the developed Competency and Readiness Evaluation website.



Paired Samples t-test

This inferential statistic is the primary tool for the One-Group Pretest-Posttest design. It will compare the significant difference between the pre-test scores and post-test scores of the same group of respondents. It will help determine if the website contributed positively to their learning performance.

If the calculated *p-value* is less than the significance level (usually 0.05), the Null Hypothesis (H_0) will be rejected, confirming the system's effectiveness.

Pearson Correlation

The attitude test will examine whether there is a relationship between the pre-service teachers' attitude or confidence level and their actual performance or readiness. The study will use the Pearson Product-Moment Correlation to measure how strongly these variables are connected. This will show if a more positive attitude or higher confidence is associated with better performance.

Interview Guide

Responses in interviews will be analyzed using Thematic Analysis, following the steps of Braun & Clarke:

1. Reading and familiarizing with responses
2. Identifying keywords and initial codes
3. Grouping similar ideas
4. Forming themes
5. Reviewing and refining the themes
6. Writing an explanation based on the patterns found

Themes will be used to support, explain, or contrast the results of the quantitative data.



REFERENCES:

- Julia, J., Subarjah, H., Maulana, M., Sujana, A., Isrokatun, I., Nugraha, D., & Rachmatin, D. (2020). Readiness and competence of new teachers for career as professional teachers in primary schools. *European Journal of Educational Research*, volume–9–2020(volume–9–issue–2–april–2020), 655–673. <https://doi.org/10.12973/eu-jer.9.2.655>
- N.d.). Researchgate.net. Retrieved July 30, 2025, from https://www.researchgate.net/publication/389164724_Readiness_of_Bachelor_of_Technical-Vocational_Teacher_Education_Pre-Service_Teachers_of_Camarines_Sur_Polytechnic_Colleges?fbclid=IwY2xjawL2yx1leHRuA2FlbQlXmABicmlkETBEOEZCNEdEaThZYW9IRU5sAR4A7vUT8ml7mhCDIXJbH7TN0cZ4rCgW5de-dM6a2_9Gy-njyz0R8KXhsPXnDg_aem_dgA7m79DT8xCJcmzKK2iNg
- Preservice Teachers' Readiness towards Integrating AI-Based Tools in Education: A TPACK Approach.
- (PDF) teachers' continuing professional development based on the Philippine Professional Standards for Teachers. (n.d.-b). https://www.researchgate.net/publication/391687879_Teachers'_Continuing_Professional_Development_Based_on_the_Philippine_Professional_Standards_for_Teachers
- (PDF) experiential learning: Experience as the source of learning and development. (n.d.). https://www.researchgate.net/publication/235701029_Experiential_Learning_Experience_As_The_Source_Of_Learning_And_Development
- *PPST Resource Package*. Teacher Education Council. (n.d.). <https://tec.deped.gov.ph/downloadable/ppst-resource-package/>
- *Subject leader*. Pedagogical Content Knowledge | Gatsby. (n.d.). <https://www.improvingtechnicaleducation.org.uk/teacher-education/your-journey/subject-leader/pedagogical-content-knowledge>
- Al-Fraihat, D., Joy, M., & Sinclair, J. (2020). Evaluating E-learning systems success using TAM. *Computers and Education*, 144, 103731.
- Bandura, A. (1997). *Self-Efficacy: The Exercise of Control*. W.H. Freeman.
- Cabanillas, L. et al. (2024). Pre-service teachers' readiness towards integrating AI tools. *Education & Information Technologies*.



- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance. *MIS Quarterly*, 13(3), 319–340.
- De Las Peñas, M. et al. (2019). Gaps in Philippine teacher education. *Philippine Journal of Teacher Education*.
- DepEd. (2017). *Philippine Professional Standards for Teachers*.
- Durak, G., & Ataizi, M. (2021). Web-based assessment platforms in teacher education. *Turkish Online Journal of Distance Education*.
- Dwivedi, Y. et al. (2023). AI in education: Opportunities and challenges. *Journal of Business Research*.
- Julia, J. et al. (2020). Readiness and competence of new teachers. *European Journal of Educational Research*, 9(2), 655–673.
- Nacario, R., & Verano, P. (2023). PPST-aligned readiness of BTVTED PSTs. *ResearchGate*.